Preference and Reward Optimization for Superconductor Crystal Diffusion

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Abstract

Crystal diffusion generators for superconductor design produce many candidates, but most are discarded by a costly post-hoc physical screening pipeline. We post-train the GuidedMatDiffusion equivariant DiffCSP generator with two complementary methods that fold cheap ML-surrogate signals into training: online Group Relative Policy Optimization with a BEE-NET and MEGNet reward, and offline Diffusion Direct Preference Optimization with CHGNet stability scores. Both methods freeze the equivariant CSPNet decoder and update only a per-node adapter pathway, preserving the symmetries of the generator. Both raise hit rate on their target screening axis, both reveal the same reward-hacking failure mode in which the policy improves its scored axis at the cost of an unscored one, and the binding constraint on this style of post-training turns out to be the surrogate ensemble rather than the optimization algorithm. Code and reproduction artifacts are available at https://github.com/jwei302/material-diffusion-RL.

1 Introduction and Problem Definition

Superconductors below a critical temperature T_c are sought for energy transmission, high-field magnets, sensors, and quantum devices, but candidate structures must simultaneously be chemically plausible, metallic, thermodynamically stable, and have favorable electron–phonon coupling. We build directly on GuidedMatDiffusion [2], which fine-tunes the DiffCSP equivariant backbone on superconductor data and conditions sampling on T_c through classifier-free guidance, with downstream screening through CrystalNN validity, MEGNet metallicity and formation-energy filters, and electron–phonon density-functional theory applied entirely after the fact. In our reproduction only a quarter of valid samples beat the SFT pool's stability quantile cutoff and fewer than three-fifths pass even the cheap filter of validity, metallicity, and negative formation energy, so most sampling compute is spent on candidates the screen rejects.

We fold the cheap end of that screen into the training signal of the generator itself. Because the underlying generator is a joint equivariant diffusion over lattice matrices, fractional coordinates, and atom-type one-hots with permutation, SE(3), and translational symmetries inherited from the DiffCSP backbone, any post-training update that breaks those symmetries cannot produce physical crystals, so both methods leave the equivariant CSPNet decoder untouched and update only a per-node adapter and property-embedding pathway that is equivariant by construction. The two methods are online Group Relative Policy Optimization with a composite BEE-NET and MEGNet reward and offline Diffusion Direct Preference Optimization with CHGNet stability scores, and the comparison between online and offline, reward-model and preference, and target-alignment and stability surrogate is what makes the joint study informative.

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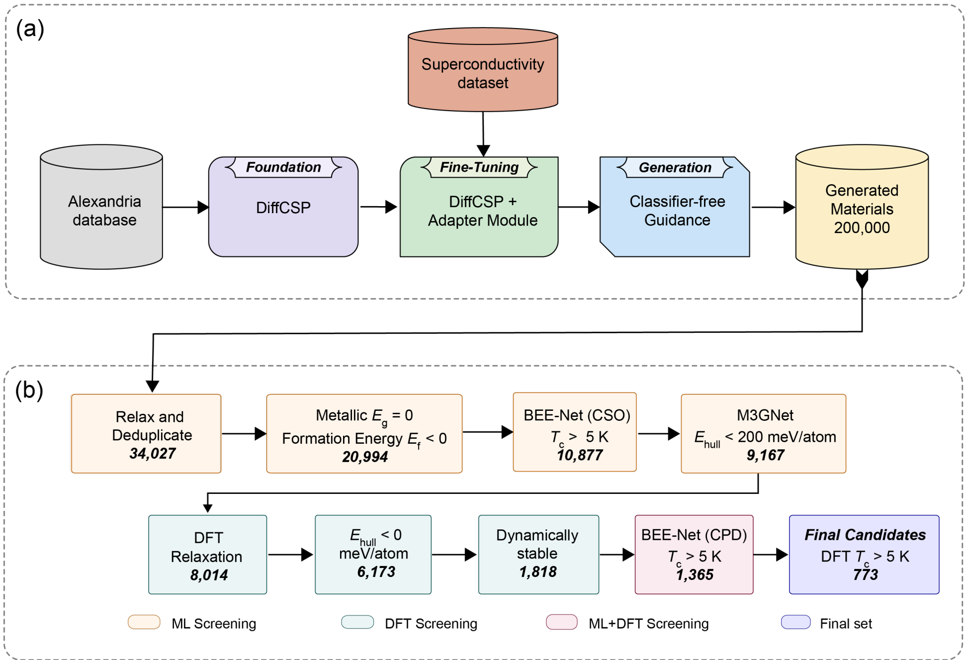

Figure 1: GuidedMatDiffusion workflow. The downstream physical screening stages on the right are applied post-hoc to samples drawn from the SFT generator. Our project folds the cheap stages of that screen back into the generator itself, by post-training the equivariant DiffCSP backbone with reward-based GRPO and preference-based DPO against ML-surrogate signals derived from the same screening pipeline.

2 Dataset

We use the processed superconductivity dataset `supccomb_12` released with GuidedMatDiffusion, which combines candidate structures and electron–phonon-derived labels from prior high-throughput studies. Each sample is a periodic crystal paired with the Allen–Dynes corrected critical temperature `tcad`. Each structure is represented as a periodic atom graph with atomic-number node attributes and CrystalNN edges under periodic boundary conditions, while the diffusion model itself operates on lattice parameters, fractional coordinates, and atom-type one-hots. The local split contains `train.csv`, `val.csv`, and `test.csv`, and the training CSV is also used to sample target prompts for GRPO. The strongly right-skewed $T_{\mathrm{c,AD}}$ distribution motivates upper-quartile target sampling for GRPO and energy-quantile pair filtering for DPO, and high-T_{c} structures are not concentrated in a single basin of the SFT feature space, so an inference-time guidance shift alone cannot raise hit rate without distortion. The full EDA table and visualizations are in Appendix B.

Statistic	**Value**
Total samples	7,183
Split sizes (train / val / test)	5,746 / 718 / 719
Max atoms per cell	12
Nodes per graph	4.91 ± 1.87
Undirected edges per graph	8.46 ± 8.67
Graph density	0.77 ± 0.19
Avg. clustering coefficient	0.45 ± 0.46
Mean node degree	2.96 ± 1.52
Unique elements per graph	2.90 ± 0.59
$T_{\mathrm{c,AD}}$ mean $\pm$ std	3.60 ± 5.22 K
$T_{\mathrm{c,AD}}$ quartiles	0.05 / 0.99 / 5.48 K
Fraction $T_{\mathrm{c,AD}} >$ 5/10/20 K	26.5% / 12.5% / 1.6%

Table 1: EDA summary for the supccomb_12 superconducting dataset.

3 Related Work

Equivariant crystal diffusion. DiffCSP [1] introduced joint equivariant diffusion over crystal lattices and fractional coordinates with an SE(3)-equivariant CSPNet decoder and a wrapped-normal coordinate score on the torus, and GuidedMatDiffusion [2] extends it with classifier-free T_c conditioning. Both works only modify sampling at inference and leave downstream screening as a post-hoc filter, so unlike them we train the adapter pathway directly against ML-surrogate screening signals while inheriting the equivariance of the frozen decoder.

Materials surrogate models. MEGNet [3] predicts band gap and formation energy, M3GNet [4] is a universal interatomic potential, BEE-NET [5] approximates electron–phonon superconductivity quantities, and CHGNet [10] is a universal pretrained energy model. These surrogates are imperfect DFT substitutes but fast enough to make post-training feasible. Prior work uses them as post-hoc screens applied to candidates already produced by a generator, while we use them as training signals whose gradients shape the generator parameters.

Preference and reward optimization. Classifier-free guidance [6] steers samples without changing parameters, DPO [7] optimizes from pairwise preferences, Diffusion-DPO [9] extends this to image diffusion under a single-component pixel loss, and GRPO [8] normalizes rewards within sampled groups so no value model is needed. Unlike Diffusion-DPO, which fine-tunes the entire image denoiser, our DPO operates over the three-component DiffCSP regression error and freezes the equivariant backbone. Unlike standard GRPO for language modeling, ours runs over branched periodic-crystal diffusion rollouts with per-step log-probabilities under the wrapped-normal coordinate score, which required custom modifications to the sampler.

4 Methodology

4.1 Base Model: GuidedMatDiffusion

The base generator is the released GuidedMatDiffusion checkpoint for `supccomb_12`. The underlying DiffCSP backbone is a joint equivariant diffusion that noises lattice matrices, fractional coordinates under a wrapped-normal score on the torus, and atom-type one-hots, with denoising performed by an SE(3)- and permutation-equivariant CSPNet decoder operating on periodic atom graphs. GuidedMatDiffusion pretrains this backbone on the Alexandria materials database and fine-tunes it on superconductor data, with property conditioning injected through a per-node adapter pathway and a global property embedding consumed at inference through classifier-free guidance. We freeze the equivariant decoder and update only the adapter and property-embedding parameters under both methods, so the lattice, coordinate, and atom-type symmetries of the SFT generator are inherited rather than relearned.

4.2 GRPO Fine-Tuning

Rollout construction. For each GRPO group we sample a target $T_{\mathrm{c,AD}}$ prompt from the training distribution with a bias toward the upper quartile and an atom count from the empirical `supccomb_12` atom-count prior. The policy first runs a shared diffusion prefix and then branches into K stochastic completions, which creates comparable candidates conditioned on the same target and prefix.

Reward model. Each branch is scored by a composite surrogate reward. BEE-NET provides dense shaping toward the requested $T_{\mathrm{c,AD}}$ and MEGNet provides terminal rewards for metallicity and formation energy:

$$\phi_{\mathrm{BEE}}(x, y) = -\frac{|\widehat{T_{\mathrm{c,AD}}}(x) - y|}{\sigma_{T_{\mathrm{c,AD}}}} + 0.25\frac{\max(\widehat{T_{\mathrm{c,AD}}}(x) - y, 0)}{\sigma_{T_{\mathrm{c,AD}}}} \tag{1}$$

$$R_{\mathrm{MEG}}(x) = 0.5\mathbf{1}\{\widehat{E}_g(x) \leq 0.05\} - 0.5\,\mathrm{softplus}\Big(\widehat{E}_{\mathrm{form}}(x)/0.05\Big)\,. \tag{2}$$

Policy objective. For each branch we recompute per-step diffusion log-probabilities under the current policy and a frozen reference policy, normalize branch rewards within the group to form advantages, and update with a clipped ratio objective regularized by reference KL:

$$\mathcal{L}_{\mathrm{GRPO}} = -\mathbb{E}_{b,t}\left[\min(r_{b,t}A_{b,t}, \mathrm{clip}(r_{b,t}, 1-\epsilon, 1+\epsilon)A_{b,t})\right] + \beta_{\mathrm{KL}}\,\mathbb{E}_{b,t}\left[\log p_\theta - \log p_{\mathrm{ref}}\right]. \quad (3)$$

The KL term keeps the policy in an equivariance-preserving neighborhood of the SFT prior and the policy decoder is structurally unchanged, so all symmetry properties of CSPNet are inherited.

4.3 DPO Fine-Tuning

Pair construction. We build a preference dataset offline by drawing two thousand samples from the SFT model at scaled y corresponding to $T_\mathrm{c} = 10$ K and $w = 2.0$, scoring each with a CHGNet single-point energy-per-atom evaluation, and pairing top-quartile and bottom-quartile samples under a reward-gap filter. For each pair we cache four timestep and noise draws and the three-component reference regression error, so training never forwards the reference policy through GPU memory.

Reward signal. For each pair we evaluate a three-term denoising error matched to the supervised diffusion loss of GuidedMatDiffusion, with one term per modality of the joint equivariant diffusion. Letting ϕ index the policy or reference,

$$\Delta_\phi(x) = w_L\,\|\varepsilon_L - \hat{\varepsilon}^L_\phi\|^2 + w_F\,\|s_F - \hat{s}^F_\phi\|^2 + w_A\,\|\varepsilon_A - \hat{\varepsilon}^A_\phi\|^2, \quad (4)$$

where s_F is the wrapped-normal score on the torus and w_L, w_F, w_A are the SFT loss weights so the DPO signal is calibrated to the same per-modality scale.

Policy objective. The Bradley–Terry preference loss applied to the implicit-reward gap between policy and reference is

$$\mathcal{L}_{\mathrm{DPO}} = -\mathbb{E}\,\log\sigma\Big(-\beta T\left[\Delta_\theta(x^w) - \Delta_{\mathrm{ref}}(x^w) - \Delta_\theta(x^l) + \Delta_{\mathrm{ref}}(x^l)\right]\Big), \quad (5)$$

where T is the total timestep count and β is the preference strength swept across five values. Only adapter and property-embedding parameters are trainable, so the equivariant CSPNet decoder is frozen and the adapter pathway is permutation-equivariant in atoms by construction. Because the adapter mixin is zero-initialized at the SFT checkpoint, the DPO loss equals exactly $\log 2$ at initialization, which we verify numerically as a unit test. Implementation contributions for both methods are listed in Appendix A.

The shared baseline for both methods is the frozen GuidedMatDiffusion `supccomb_12` checkpoint sampled under the protocol on which the post-trained policy is evaluated. All reported runs use seed zero because of compute budget, and runs are deterministic up to GPU-kernel non-determinism. Hyperparameter tables and hardware details are in Appendix E.

5 GRPO Experiments and Results

Setup. GRPO is evaluated on a thirty-two-prompt target-temperature evaluation comparing the policy against the frozen reference at guidance weights $\{0.5, 1.0, 2.0\}$. We report mean absolute $T_{\mathrm{c,AD}}$ error and uplift, metallic pass rate, negative-formation-energy pass rate, conjunctive cheap-filter pass rate, invalid rate, and duplicate rate. The metrics target the GRPO screening axis of target-temperature alignment and metallicity.

Quantitative results. Table 2 shows the final 150-update GRPO run. The GRPO policy's best deployment guide weight was 2.0, while the reference's best guide weight was 0.5 under the composite selection rule.

The full guidance-weight ablation behind this best-weight selection is reported in Appendix D.

Analysis. GRPO improves target-temperature alignment and metallicity while preserving validity and diversity, but the unrewarded formation-energy pass rate falls from above eight-tenths to about half. The BEE-NET-and-MEGNet composite contains no stability term, so the policy learns it can move into formation-energy-disfavored regions of structure space without paying a cost, which is the canonical reward-hacking pattern under an under-specified surrogate.

Metric	GRPO policy	Frozen reference	Delta
Best guide weight	2.0	0.5	–
Mean absolute $T_{c,\mathrm{AD}}$ error (K)	4.821	5.061	-0.241
Mean $T_{c,\mathrm{AD}}$ uplift (K)	-1.709	-4.239	$+2.530$
Metallic pass rate	0.906	0.750	$+0.156$
Negative E_{form} pass rate	0.531	0.812	-0.281
Cheap-filter pass rate	0.500	0.594	-0.094
Invalid rate	0.000	0.000	0.000
Duplicate rate	0.000	0.000	0.000
Composite score	-0.503	-0.576	$+0.073$

Table 2: Final GRPO report-run metrics at update 150, compared with the frozen reference policy on the same 32-prompt evaluation protocol.

Pool	N valid	Med. E/atom (eV)	Hit-rate@25	Top-500 med. (eV)	# chemsys
SFT baseline	1999	-9.85	25.0%	-10.96	826
DPO $\beta = 25$	1998	-9.93	27.1%	-10.95	830
DPO $\beta = 5$	1998	-9.95	27.7%	-11.01	833
DPO $\beta = 1$	**1998**	$\mathbf{-9.99}$	**30.2%**	$\mathbf{-11.08}$	**797**
DPO $\beta = 0.5$	1996	-10.11	33.7%	-11.21	791
DPO $\beta = 0.1$ †	1999	-11.83	90.2%	-12.40	**158**

Table 3: DPO five-point β sweep, with two thousand samples per pool drawn at the same conditioning and seed as the SFT baseline and scored by CHGNet single-point energy per atom. Hit-rate@25 is the fraction of valid samples below the SFT pool's twenty-fifth-percentile energy cutoff. The honest sweet spot of the sweep is $\beta = 1$. † $\beta = 0.1$ is reward-hacked, with chemistry collapsing onto Ta- and Re-rich five-d transition metals as documented in Appendix C.

6 DPO Experiments and Results

Setup. DPO is evaluated on a two-thousand-sample pool drawn at $w = 2.0$ and scaled y corresponding to $T_c = 10$ K, with a four-times-compute SFT best-of-N control to separate genuine distribution shift from screening at larger sample volume. We report median CHGNet energy-per-atom, hit rate at the SFT pool's twenty-fifth-percentile cutoff, top-five-hundred median energy, and unique chemical-system count. The metrics target the DPO screening axis of thermodynamic stability proxy.

Quantitative results. The DPO sweep is summarized in Table 3, with full β-sweep figures, energy-distribution shifts, chemistry audits, training dynamics, and the compute-matched control in Appendix C.

Stability preferences raise hit rate. Median and top-K energies move downward monotonically as β decreases from twenty-five toward one, and the hit rate at the SFT twenty-fifth-percentile cutoff rises from a quarter to roughly three-tenths at $\beta = 1$ while unique-chemsys count is preserved. Small consistent preference signal is enough to bias the zero-initialized adapter mixin toward structurally lower-energy pockets that the SFT generator already covers, without re-exploring them through expensive online rollouts.

Aggressive β collapses chemistry. At $\beta = 0.1$ the apparent hit rate jumps to roughly nine-tenths but unique-chemsys count collapses by a factor of five, with the bulk of the pool falling into ten Ta- and Re-rich five-d transition-metal chemsys that CHGNet under-scores relative to higher-fidelity surrogates. A low β reduces the implicit Kullback–Leibler regularization toward the SFT prior, freeing the adapter to push toward a basin where the surrogate's calibration error is largest, so the metric jump at $\beta = 0.1$ is surrogate exploitation rather than genuine stability gain, and the honest sweet spot is $\beta = 1$.

Compute-matched comparison. At matched compute the DPO pool dominates an SFT best-of-N pool on top-K screening yield, but a four-times-compute SFT pool dominates the one-times-compute DPO pool by sheer volume. DPO durably shifts the sampling distribution so every sample carries the post-training advantage, while additional SFT samples are independent draws from a fixed distribution, so the practically interesting comparison of DPO at four-times compute against SFT at four-times compute is left to follow-on work.

7 Discussion, Conclusion, and Limitations

Both pipelines move the generator measurably in the intended direction without immediate collapse. GRPO improves target-temperature alignment and pushes metallic pass rate up by more than ten percentage points, and DPO at $\beta = 1$ raises the energy-quantile hit rate from a quarter to roughly three-tenths while preserving chemistry diversity. The shared intuition is that surrogate signal is most useful as a small consistent push on the sampling distribution rather than a target the policy must aggressively saturate, because the post-trained generator inherits the SFT prior's coverage and post-training simply tilts that prior in the desired direction.

The two methods fail in the same characteristic way despite differing in algorithm and surrogate, which is the most informative finding of the comparison. GRPO improves the rewarded axes of target-temperature and metallicity while the unrewarded formation-energy pass rate falls from above eight-tenths to about half, because its composite contains no stability term. DPO at $\beta = 0.1$ collapses onto Ta- and Re-rich five-d transition metals, because CHGNet single-point energy is the only surrogate and its calibration is biased on those systems. In both cases the policy is doing exactly what the loss asks of it and the loss is asking the wrong question, so the binding constraint on surrogate-based post-training in this regime is the composition of the surrogate ensemble rather than the choice of optimization algorithm. A practical contrast is that GRPO forwards surrogate reward models in the training loop while DPO pre-caches them once at pair-construction time, so the DPO five-point sweep finishes in roughly twenty minutes on an H200 while a single GRPO run at one hundred fifty updates takes about ninety minutes. Offline preference optimization is more compute-efficient at the sweep level when the screen is fixed, while online reward optimization is the right tool when the screen evolves or when reward-landscape exploration matters.

The main limitations are that our surrogate ensembles are deliberately narrow, that DPO sees only stability while GRPO sees only target-temperature and metallicity, that all reported runs use a single random seed because of compute budget, and that no run validates top candidates with density-functional theory. The natural follow-on work is to combine the two methods, with GRPO reward composites that include CHGNet or M3GNet stability terms and DPO preference pairs that score relaxation stability rather than single-point energy, both of which are outside our class-project compute budget. Code organization, environment pins, and full reproduction commands are in Appendix F.

References

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A Implementation contributions

GRPO. Wenhe added the GRPO entry point, prompt sampler, reward manager, and trainer in `diffcsp/run_rl.py` and `diffcsp/rl/`, modified the diffusion sampler to expose shared-prefix branch rollouts and per-step log-probabilities, integrated BEE-NET and MEGNet smoke tests, fixed nested Hydra loading for checkpoint configs, added dtype hardening after BEE-NET changed the global Torch default dtype, clamped wrapped-normal log-probabilities to avoid NaNs at deterministic final diffusion steps, and added `train_progress.jsonl` logging for reproducibility.

DPO. Jeffrey implemented `DPODiffusion` in `diffcsp/pl_modules/dpo_module.py` with the three-component MSE under cached noise and timestep, `PreferenceDataset` and `collate_pairs` in `diffcsp/pl_data/preference_dataset.py` which enforces shared t, m within a batch so the policy decoder evaluates a single timestep per forward pass, the offline pair-builder under `scripts/dpo/build_preference_pairs.py` that runs the reference policy once per pair and caches its denoising errors, the CHGNet score-cache keyed by canonical structure hash, the chemsys-and-uniqueness audit pipeline that flags reward-hacked pools, and a unit test for the log-2 starting-point sanity check and antisymmetry of the DPO logit.

B Dataset EDA

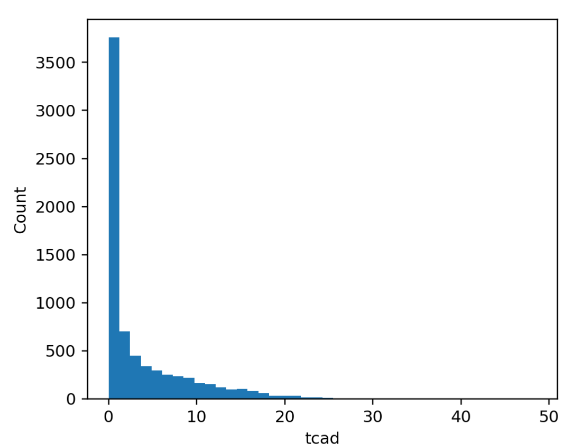

Figure 2: $T_{\mathrm{c,AD}}$ distribution of the supccomb_12 training set. The strongly right-skewed shape motivates upper-quartile target sampling for GRPO and energy-quantile pair filtering for DPO.

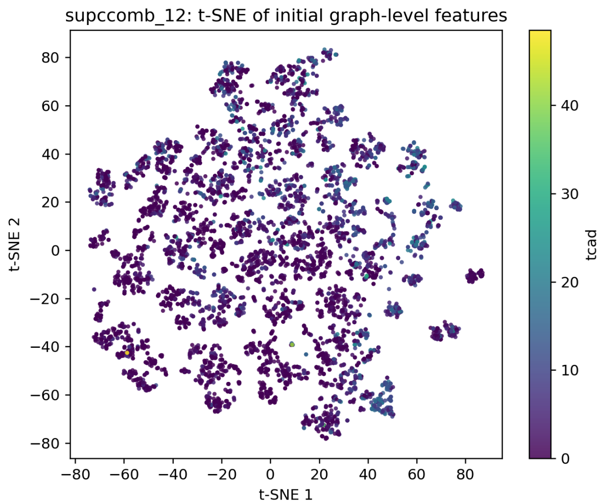

Figure 3: t-SNE projection of CSPNet initial hidden features colored by $T_{\mathrm{c,AD}}$. High-T_{c} structures are not concentrated in a single basin of the SFT feature space, which is why a parameter-space update through post-training is the natural alternative to inference-time guidance.

C DPO post-training visualizations

We provide visualizations characterizing the quality and failure modes of DPO post-training. Figure 4 sweeps β against the four metrics from Table 3, Figure 5 stacks per-pool energy distributions to expose both the directional shift at $\beta = 1$ and the heavy-tail collapse at $\beta = 0.1$, Figure 6 flags the $\beta = 0.1$ chemistry collapse, Figure 7 reports training dynamics, and Figure 8 reports the compute-matched control.

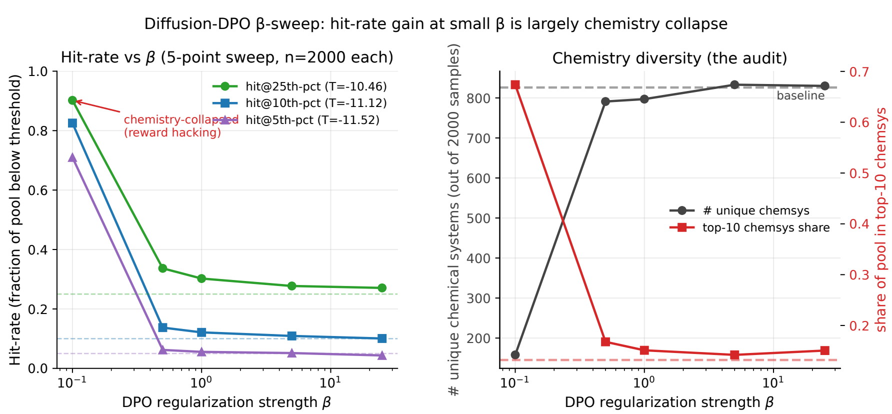

Figure 4: Five-point β-sweep characterization for the DPO pipeline. Lower energy is better. The honest sweet spot is $\beta = 1$, where hit rate rises while chemsys diversity is preserved.

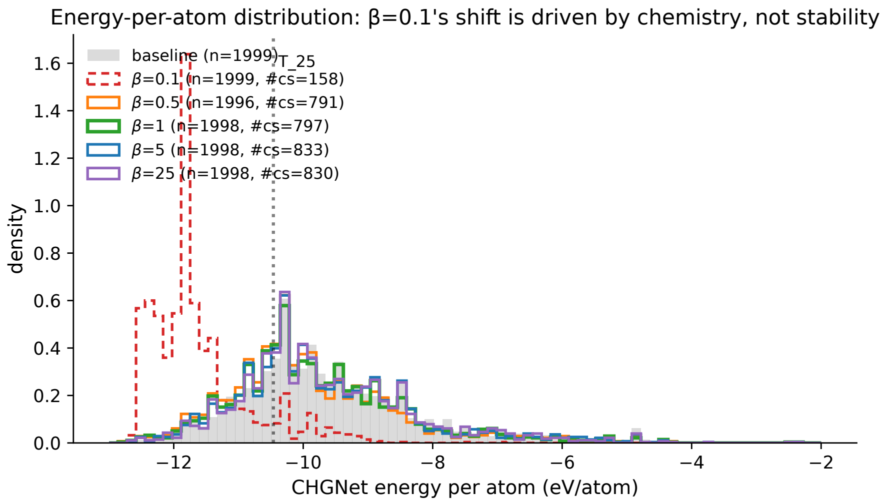

Figure 5: Per-pool CHGNet energy-per-atom distributions. The $\beta = 1$ pool shifts cleanly downward relative to SFT, while the $\beta = 0.1$ pool develops a heavy left tail that signals reward-hacking.

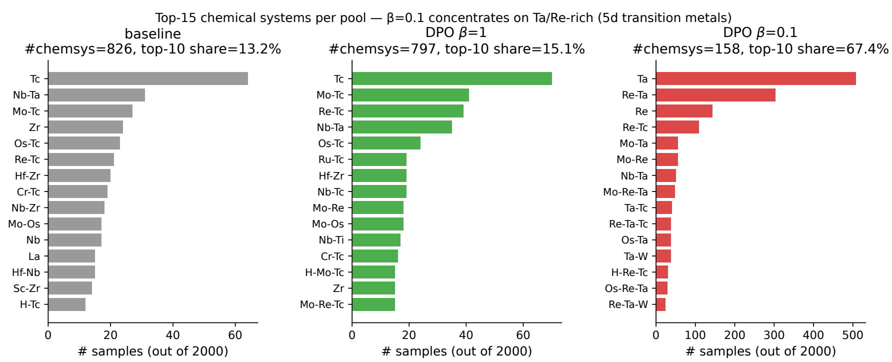

Figure 6: Chemistry-diversity audit. Unique chemsys count and top-ten chemsys share are stable for $\beta \in \{1, 5, 25\}$ and collapse sharply at $\beta = 0.1$.

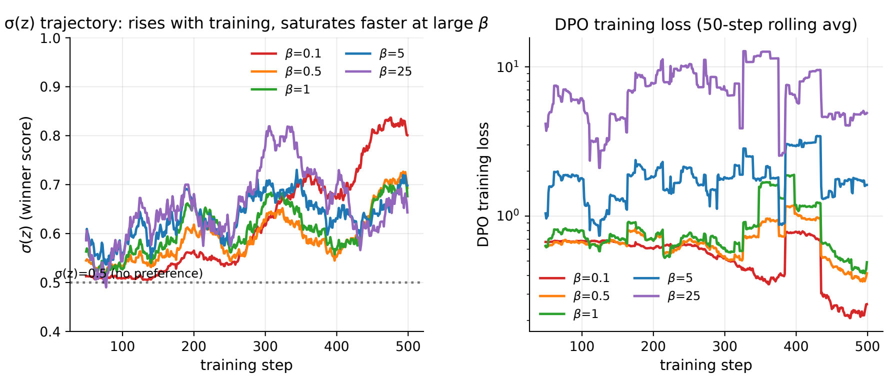

Figure 7: Trajectory of $\sigma(z)$ across the β sweep. Larger β saturates the preference logit faster while smaller β keeps the loss landscape softer.

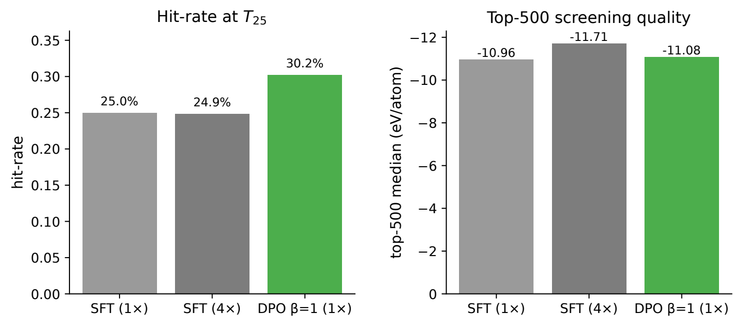

Figure 8: Compute-matched best-of-N control. DPO at one-times compute beats SFT at one-times compute, but four-times-compute SFT screening recovers more top-K candidates by volume.

D GRPO guidance-weight ablation

Model	w	Abs. err.	Uplift	Metal	Neg. E_{form}	Cheap	Invalid
GRPO	0.5	5.242	−3.411	0.594	0.688	0.312	0.000
GRPO	1.0	5.341	−3.729	0.750	0.906	0.656	0.000
GRPO	2.0	**4.821**	−1.709	**0.906**	0.531	0.500	0.000
Reference	0.5	5.061	−4.239	0.750	0.812	0.594	0.000
Reference	1.0	5.323	−3.338	0.781	0.562	0.375	0.000
Reference	2.0	6.069	−2.805	0.812	0.656	0.500	0.000

Table 4: Guidance-weight ablation underlying the GRPO headline result in Table 2.

E Hyperparameters and compute

Experiments ran on a cluster node with an NVIDIA H200 GPU. BEE-NET and CHGNet ran on CUDA, MEGNet used a TensorFlow CPU fallback because its CUDA 11 runtime libraries were unavailable in our environment. The final GRPO report run took roughly ninety minutes including evaluation, and the entire DPO five-point β sweep finished in under twenty minutes because reference denoising errors are pre-cached.

Table 5: GRPO hyperparameters for the 150-update report run.

Hyperparameter	Value
Maximum updates	150
Groups per update	1
Branches per group	2
Diffusion prefix step	50
Reward interval	25
Adam learning rate	5×10^{-7}
PPO clip ϵ	0.02
KL coefficient β_{KL}	0.05
Gradient norm clip	1.0
BEE-NET ensemble size	1
M3GNet / proxy rewards enabled	no
Eval prompts	32

Table 6: DPO hyperparameters for the five-point β sweep.

Hyperparameter	Value
Pair pool size	1000
Timestep draws per pair	4
Top / bottom fraction for pairs	0.25 / 0.25
Reward gap margin	0.5
Diffusion total timesteps T	1000
Component weights w_L, w_F, w_A	$1, 1, 20$
β values	$\{0.1, 0.5, 1, 5, 25\}$
Adam learning rate	1×10^{-4}
Batch size	1
Maximum steps	5000
Validation interval	500
Trainable parameter fraction	$\approx 14\%$

F Code and reproduction

Core GRPO files are in `diffcsp/rl/` and `diffcsp/run_rl.py`, the DPO module and preference dataset are `diffcsp/pl_modules/dpo_module.py` and `diffcsp/pl_data/preference_dataset.py`, and DPO entry points are under `scripts/dpo/`. GRPO uses the `matdiff39` environment with BEE-NET and MEGNet, and DPO uses a CHGNet-augmented uv environment whose pinned dependencies are committed in `pyproject.toml` and `uv.lock`. Reproduction artifacts include the cached preference pairs, all CHGNet score caches, per-β training-metric CSVs, and final figures.

GRPO.

```
python -m diffcsp.run_rl \
  expname=rl_grpo_report_150 \
  paths.policy_checkpoint=$PROJECT_ROOT/hf_models/superconductor_generator \
  grpo.schedule.max_updates=150 \
  grpo.sampling.num_groups_per_update=1 \
  grpo.sampling.num_branches=2 \
  grpo.sampling.prefix_t=50 \
  grpo.sampling.reward_interval=25 \
  grpo.eval.num_prompts=32 \
  grpo.rewards.bee.ensemble_size=1 \
  grpo.rewards.bee.device=cuda \
  grpo.rewards.m3gnet.enabled=false
```

DPO.

```
python scripts/dpo/build_preference_pairs.py \
  --score_file dpo_artifacts/day1_baseline_pool/scores_y10k.json \
  --cif_dir   dpo_artifacts/day1_baseline_pool/cifs_y10k \
  --model_path models/superconductor_generator \
  --out_path  dpo_artifacts/pairs_v1.pt \
  --top_frac 0.25 --bottom_frac 0.25 --gap_margin 0.5 \
  --k_draws 4 --max_pairs 1000 --y 1.2218

python scripts/dpo/dpo_train.py \
  --pairs_file dpo_artifacts/pairs_v1.pt \
  --model_path models/superconductor_generator \
  --out_dir   dpo_artifacts/dpo_b1 \
  --beta 1 --max_steps 5000 --batch_size 1 --gpus 1

python scripts/dpo/sample_dpo_ckpt.py \
  --ref_model_path models/superconductor_generator \
```

```
  --dpo_ckpt dpo_artifacts/dpo_b1/dpo_final.ckpt \
  --save_path dpo_artifacts/eval/cifs_dpo_b1 \
  --band_gap 1.2218 --guide_w 2.0 \
  --batch_size 100 --num_batches_to_samples 20 --seed 0

python scripts/dpo/score_ehull.py \
  --cif_dir dpo_artifacts/eval/cifs_dpo_b1 \
  --out    dpo_artifacts/eval/scores_dpo_b1.json \
  --cache  dpo_artifacts/score_cache/chgnet_cache.json
```